25900248

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Core Course (DSC) - MG3DSCBTG200 - BIOPHYSICS AND INSTRUMENTATION

(2024 ADMISSION ONWARDS)

SCHEME

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

One word or sentence-Answer any 10 questions

Each question carries 2 marks

1. Give the name for a common Adsorbent? [K] / [CO1]
   Any one (Silica gel/Alumina gel etc)- 2 marks
2. Recall two methods of paper chromatography. [K] / [CO3]
   Ascending and Descending paper chromatography- 2 marks
3. State the principle behind dialysis. [U] / [CO3]
   Principle- 2 marks
4. Enumerate any two chemical properties of water. [K] / [CO1]
   Any two chemical properties- 2 marks
5. Recall which amino acid shows restricted angles in Ramachandran plot. [K] / [CO2]
   Proline- 2 marks
6. Name one isotope used in autoradiography. [K] / [CO4]
   Any one isotope used in autoradiography-2 marks
7. Identify the type of molecules separated by Gel Permeation Chromatography.[U] / [CO3]
   Proteins/Nucleic acids/Biopolymers-2 marks
8. State the zeroth law of thermodynamics. [K] / [CO1]
   State the Law- 2 marks
9. Comment on the primary advantage of using HPLC over Gas chromatography? [U] / [CO3]
   Non-volatile/Thermally unstable/High molecular weight- 2 marks
10. What is required in electron microscopy to prevent scattering of electrons? [K] / [CO4]
    Vacuum/electromagnets- 2 marks

11. Name the source of IR radiation. [K] / [CO4]
    Any one source-2 marks
12. State why GC-rich DNA has a higher melting temperature. [U] / [CO2]
    Presence of three Hydrogen bonds in GC pairing- 2 marks

[2x10 = 20]

Part B

Short Essay-Answer any 5 questions

Each question carries 4 marks

13. Describe the tertiary structure of protein and list the types of bonds that stabilize it. [U] / [CO2]
    Tertiary structure of protein-explain- 2 marks, explain the types of bonds- 2 marks
14. Explain enthalpy and entropy including examples of its application in biology. [K] / [CO1]
    Definition of enthalpy-1 mark, equation- .5 mark, Definition of entropy-1 mark, equation- .5 mark, Any two examples- 1 mark
15. Recall the advantage of PAGE over AGE. [K] / [CO3]
    PAGE and AGE- expansion- 1 mark, any three advantages- 3 marks
16. State the difference between dispersion and scattering (any 4 points). [K] / [CO1]
    Dispersion – definition, example- 1 mark, scattering- definition, example- 1 mark, Any two differences- 2 marks
17. Explain the structure of four-stranded DNA (G-quadruplex) with the help of a diagram. [U] / [CO2]
    Four-stranded DNA (G-quadruplex)- 2 marks, Diagram- 2 marks
18. Define and explain the concept of sedimentation coefficient and its unit [K] / [CO3]
    Sedimentation coefficient-Definition-1 mark, explanation-2 marks, unit- 1 mark
19. Mention and explain four applications of absorption in daily life. [K] / [CO1]
    Any four applications of absorption in daily life (Physical and biological)-4 marks

[4x5 = 20]

Part C

Essay type questions-Answer any 1 questions

Each question carries 10 marks

20. Define a biosensor and outline its principle and basic working mechanism. [K] / [CO4]Biosensor-Definition, example- 2 marks, Principle- 3 marks, Working mechanism -explain with diagram- 5 marks
21. Explain in detail the concept of diffusion and importance in biological context. [K] / [CO1]
    Diffusion- definition- 2 marks, Principle- 3 marks, Applications (5points) – 5 marks

[10x1 = 10]